

Question bank

NEBOSH Process Safety Management

Q1 process safety differs from personal safety because it protects

- a) Public only and all industries.**
- b) Public only and process industries.**
- c) Workers only and all industries .**
- d) Workers, public and process industries .**

Q2 which of the following is likely to be most helpful in developing a positive health and safety culture ?

- a) Senior management creating workforce fear of disciplinary proceedings.**
- b) High level communication chains with diluted messages at lower levels.**
- c) Promotion of process safety leadership and board level visibility of site visits.**
- d) Management focus on productivity not unreasonable safety regulation.**

Q3 An example of 'organisational learning' within process safety management is documented

- a) Systematic critical examination of a safety management system.**
- b) Statistical information for a specific accident type within the oil and gas industry.**
- c) Management processes ensuring retention of process safety corporate knowledge.**
- d) Records of a worker's skills, knowledge and behaviour development.**

Q4 The root cause of insufficient process safety resources, such as engineering support, is inadequate process safety

- a) design .**
- b) planning.**
- c) operation .**
- d) leadership.**

Q5 which of the following is most likely to help you develop a safe system of work ?

- a) Task analysis
- b) Data analysis
- c) Causal analysis
- d) Statistical analysis

Q6 What is normally included in the periodic review of the safety performance of contractors working on site?

- a) A request for method statements, risk assessments and official licenses
- b) Scrutiny of contractor health and safety policy and previous prosecutions
- c) Analysis of contractor training records, qualifications and past experience
- d) Inspections to check compliance and accident data review meetings

Q7 Which of the following risk management tools identifies deviations from normal activity and applies guide words to process variables?

- a) A hazard identification study
- b) An event tree analysis
- c) A failure mode effects analysis
- d) A hazard and operability study

Q8 Inspecting a workplace where contract workers are carrying out work is

- a) actively monitoring.
- b) reactively monitoring.
- c) condition monitoring.
- d) performance monitoring.

Q9 A permit-to-work is usually required in preparation for an activity with

- a) any risk.
- b) no risk.
- c) high risk.
- d) low risk.

Q10 Maintenance of bunds primarily contributes to

- a) operational excellence.
- b) change management.
- c) emergency preparedness.
- d) inventory management.

Q11 Which of the following is the best indicator of the state of process safety on a plant?

- a) The number of lost time injuries
- b) The number of hours since last injury
- c) The number of loss of containment incidents
- d) The rate of sickness absence

Q12 which of the following would be the highest priority to discuss during shift handover ?

- a) Communication radio in good working order
- b) Plant log book signed by outgoing operator
- c) Emergency shutdown systems on bypass
- d) Safe operating envelope parameters in alarm

Q13 Asset management includes consideration of integrity standards, such as maintainability and ease of inspection at

- a) design stage.
- b) construction stage.
- c) operation stage.
- d) modification stage.

Q14 Tank level gauges can be made more reliable through

- a) testing and maintenance.
- b) information and instruction.
- c) auditing and reporting.
- d) assessment and controls.

Q15 A consequences of a thermal runaway reaction is catastrophic plant failure through

- a) Overrunning.
- b) Over-pressure.
- c) Overloading.
- d) Overriding.

Q16 safe operating envelopes involve adhering to

- a) upper and lower operating limits.
- b) upper operating limits only.
- c) variable operating limits.
- d) lower operating limits only.

Q17 Steam that does not wet the surface it touches is usually called

- a) saturated.
- b) high pressure.
- c) superheated.
- d) Vacuum.

Q18 Overfilling of tanks can be prevented most effectively through a combination of

- a) Calculation of mass flow rates in, knowledge of tank capacity and low level alarms.
- b) Calculation of mass flow rates in and mass flow rates out , knowledge of tank capacity and low level alarms.
- c) Calculation of mass flow rates in and mass flow rates out , knowledge of tank capacity and high level alarms.
- d) Calculation of mass flow rates out , knowledge of tank capacity and high level alarms .

Q19 Shutdown can be planned , unplanned , emergency, staged or delayed.

In which of the following circumstances would a delayed process shutdown be appropriate ?

- a) An alarm indicates that the process safe operating envelope is about to be breached
- b) The power to part of the process has suddenly failed unexpectedly
- c) The cooling for a reaction vessel has been lost due to a major water leak
- d) Minor valve leaks appear two weeks before a planned shutdown

Q20 The highest risk associated with the path of the electrical current through the human body is

- a) unconsciousness.
- b) entry and exit burns.
- c) tingling where contact occurs.
- d) stopping the heart.

Q21 One of the hazards of using nitrogen as an inerting gas is

- a) increased oxygen concentrations.
- b) decreased flammability.
- c) increased flammability.
- d) decreased oxygen concentrations.

Q22 Safety critical performance standards for safety critical elements can best be derived from

- a) information from reporting and recording documents.
- b) condition monitoring within planned maintenance.
- c) analysis of the design through process hazard analysis.
- d) process operation parameters within operating procedures.

Q23 Rapid temperature change can cause cracking of materials.

This type of failure mode is known as

- a) creep.
- b) brittle fracture.
- c) thermal shock.
- d) Stress corrosion cracking.

Q24 An endothermic reaction is one that

- a) absorbs energy from its surroundings with a corresponding temperature decrease.
- b) releases energy with a corresponding temperature increase.
- c) releases energy with a corresponding temperature decrease.
- d) Absorbs energy from its surroundings with a corresponding temperature increase.

Q25 Pressure, temperature, flow, level, pH and density are common

- a) Processes associated with safe operating envelopes.
- b) Problems associated with safe operating envelopes.
- c) Parameters associated with safe operating envelopes.
- d) Procedures associated with safe operating envelopes.

Q26 Which of the appropriate type of fire extinguisher to use on metal fires?

- a) Water.
- b) Foam.
- c) Dry powder.
- d) Carbon dioxide.

Q27 Which of the following is an example of a condition needed for a dust explosion to occur?

- a) Dust concentration below lower explosive limit
- b) Dust is confined in a hopper or silo
- c) Dust particle size is very large
- d) Very high humidity

Q28 For which operating environment is ATEX- approved equipment essential?

- a) Corrosive atmospheres in offshore oil and gas installations
- b) Toxic atmospheres within confined space work activities
- c) Pressurised atmospheres requiring associated relief valves
- d) Potentially flammable atmospheres from dust or vapour

Q29 Hot gases from fires naturally rise through voids and ducts.

This behaviour is known as

- a) convection.**
- b) direct burning.**
- c) radiation.**
- d) conduction.**

Q30 For which category of worker is a Personal Emergency Plan most appropriate ?

- a) Plant workers**
- b) Vulnerable workers**
- c) Contract workers**
- d) Site workers**

Q31 A bulk fuel tank is normally filled automatically from a pipeline through a filling valve. The tank is fitted with a level gauge that enables operators to remotely monitor the level from a control room. The tank is also fitted with a high level alarm and an independent high level switch that, if activated, closes the filling valve. The filling valve can also be closed using an emergency switch located in the control room. On one occasion all of these devices failed and the tank overflowed.

While all of these devices help prevent overfilling, which is the most safety critical item required to effectively prevent the tank overfilling ?

- a) High level alarm**
- b) Level gauge**
- c) Independent high level switch**
- d) The filling valve**

Q32 You need to develop a set of Standard Operating Procedures (SOPs) to ensure the safe operation of a petroleum (gasoline) storage tank. There have been a number of incidents in the past where front line operators have failed to follow the SOPs .

Which ONE of the following is most likely to ensure that operators are aware of the SOPs , understand their purpose and follow them ?

- a) Send an email to all employees making clear that all SOPs will be complied with (non-compliance is not an option) ; include a copy of the SOPs for ease of reference .**
- b) Involve the front line operators when designing the SOPs , along with other stakeholders (eg representatives of the organisation that designed and manufactured the storage tank)**

- c) Design and display a number posters that advertise that the SOPs are in place, that they must be followed and failure to do so will result in dismissal
- d) Train the front line operators in the safe use of the gasoline storage tank, highlighting that any deviance from the SOPs will not be tolerated and will result in severe consequences

Q33 A large chemical mix tank was fitted with a refrigeration system, designed to cool the reaction mixture during a process. However, the refrigeration system was not operational because the refrigerant had been removed and used in another part of the chemical plant. this mix tank was not used very often, but was now required to produce a new batch of product for a customer. On this occasion, the reaction became a thermal runaway and operators did not know how to stop it before the tank exploded.

Removal of the refrigerant unit from process plant operation was mainly a failure to recognise the

- a) lack of competence.
- b) need for resources.
- c) implications of change
- d) operating procedure.

Q34 As a supervisor you have noticed that ,despite clear organisation guidelines, operators regularly bypass the standard operating procedures. An independent review has suggested that this is due to management failure.

Which of the following management actions is most likely to improve process safety on site?

- a) Reminder sent to all the staff highlighting the importance of process safety and that non-compliance will not be tolerated
- b) Conduct a Hazard and Operability study, recording all significant findings and actions
- c) Develop an effective competence management system, based on task analysis
- d) Conduct further investigation into the causes to verify the review findings before deciding on any further action

Q35 As a manager of a manufacturing site you have concerns that operations often suffer from fatigue when they are at work. On a number of occasions

employees have been found sleeping when in charge of manufacturing process. The process is continuous and operates over two shifts.

Which ONE of the following is most likely to ensure that safety critical information (eg details of maintenance activity from the previous shift). The process is continuous and operates over two shifts.

- a) Install cameras that monitor operators so that when they are caught asleep they can verbally warned and formally disciplined
- b) Provide members of both shifts with 15-minutes of paid time to discuss safety critical activities that have taken place during the previous shift
- c) The out-going shift leaves a note (eg in a log book) for the next shift detailing any maintenance work that is needed on process equipment
- d) Introduce a health and wellbeing programme that educates shift workers on the dangers of fatigue and what action they can take to keep alert

Q36 You are working on a site that handles combustible dusts. You have already taken precautions to reduce the chances of a primary dust explosion. However, you are worried about the risks of a secondary dust explosion.

Which is the best way to reduce this risk?

- a) Avoid ignition sources
- b) Explosion relief
- c) Explosion suppression
- d) Good housekeeping

Q37 A tank containing liquefied petroleum gas became engulfed in fire from a nearby warehouse. After a short time, the tank exploded causing damage over a large area.

What is the most likely type of event that caused this damage?

- a) A boiling liquid expanding vapour explosion
- b) A jet fire
- c) A pool fire
- d) A confined vapour cloud explosion

Q38 The site that you work on has a series of reactors in a line, one of which is found to be leaking. To keep production going, it is decided that a bypass pipe needs to be installed on the leaking reactor to keep production going.

What is the most important thing you should do before installing the bypass pipe?

- a) Review implications of the proposed change
- b) Consult the relevant international standard
- c) Develop a preventive maintenance programme

- d) Use the correct grade of steel for the process

Q39 You are conducting a hazardous area classification for a petroleum fuel storage site; in particular you are concerned with the area around the fuel pump and pipelines connected to each storage tank.

Assuming all components are in good condition, what zone is the area around the fuel pump likely to be under normal operation?

- a) Zone 21
- b) Zone 22
- c) Zone 1
- d) Zone 2

Q40 Following a tank overflow, a flammable vapour within explosive limits travelled some distance across an open roadway, contacted an ignition source and exploded.

What type of explosion is this likely to be?

- a) A confined vapour cloud explosion
- b) A secondary explosion
- c) A boiling liquid expanding vapour explosion
- d) An unconfined vapour cloud explosion

Q41 What is the main reason for sharing lessons learnt from major incident with other organisations ?

- a) Integration of quality , environmental and health and safety management systems
- b) Potential benefits are much greater for the process safety industry as a whole
- c) Accident investigations are more effective in determining immediate and root causes
- d) Operating procedures are standardised so that workers have a consistent approach

Q42 An example of 'organisational learning' within process safety management is documented

- a) Systematic critical examination of a safety management system.
- b) Statistical information for a specific accident type within the oil and gas industry.

- c) Management process ensuring retention of process safety corporate knowledge.
- d) Records of a worker's skills, knowledge and behavior development.

Q43 Process safety involves a commitment continuous

- a) impartiality.
- b) Improvement.
- c) Improvisation.
- d) Implementation

Q44 The isolation of energy sources, or equipment as part of a permit-to-work is known as

- a) lock out, tag on.
- b) lock on, tag off.
- c) lock out, tag out.
- d) lock on, tag on.

Q45 'Bowtie' is a risk management tool that identifies

- a) barriers to change.
- b) prevention barriers only.
- c) mitigation barriers only.
- d) prevention and mitigation barriers

Q46 Inspection, aging, life extension and obsolescence are all things that need to be considered as part of a plant's

- a) Start-up process.
- b) Shutdown process.
- c) Operational procedures.
- d) Maintenance strategy.

Q47 Workers carrying out shift handover require technical knowledge and expertise.

What other skill area is essential for these workers?

- a) Communications
- b) Negotiation

- c) Supervision
- d) Motivation

Q48 The concept of risk being as low as reasonably practicable is best described as the risk is as low as it can be

- a) using current technology and knowledge .
- b) taking account of the resources of the organisation .
- c) without being disproportionately costly.
- d) taking account of the priorities of the organisation

Q49 Safety critical performance standards for safety critical elements can best be derived from

- a) information from reporting and recording documents.
- b) condition monitoring within planned maintenance.
- c) analysis of the design through process hazard analysis.
- d) process operation parameters within operating procedures.

Q50 When filling a vessel with a flammable liquid, one of the main factors that increases the generation of an electrostatic is

- a) overfilling.
- b) splash filling.
- c) bottom filling.
- d) continuous filling.

Q51 process safety, performance standards are most important for

- a) safety critical systems and items of equipment.
- b) quality of standards operating procedures.
- c) quality of risk assessment procedures.
- d) safety training and competence levels.

Q52 What is the main purpose of a pressure relief valve fitted to a vessel?

- a) prevent vessel reaction
- b) prevent vessel collapse
- c) prevent thermal runaway
- d) prevent vessel rupture

Q53 What is the ideal temperature for legionella bacteria to grow?

- a) 27°C
- b) 37°C
- c) 47°C
- d) 57°C

Q54 Dividing a building into fire-resisting compartments to limit the spread of smoke and fire is an example of which type of fire precaution ?

- a) Active fire protection
- b) Fire detection system
- c) Passive fire protection
- d) Smoke detection system

Q55 One of the main requirements for a primary dust explosion to occur is a combustible dust that is

- a) airborne and within explosive limits.
- b) airborne and outside explosive limits.
- c) on a hot flat surface outside explosive limits.
- d) on a hot flat surface within explosive limits.

Q56 The transfer of heat through solids is known as

- a) convection.
- b) direct burning.
- c) radiation.
- d) Conduction.

Q57 Sometimes a 'table-top exercise' is used to simulated an emergency.

What are the main benefits of such exercises?

- a) Validates plans, develops staff competency by practising roles and tests procedures
- b) Identifies hazards, considers those affected, evaluates risk and implements suitable controls
- c) Analyses unsafe working conditions and behaviours and implements corrective actions
- d) Examines a process safety management system, identifies problems and provides solutions

Q58 You are conducting a hazardous area classification for a petroleum fuel storage site in particular you are concerned with the area around the fuel pump and pipelines connected to each storage tank.

Assuming all components are in good condition, what zone is the area around the fuel pump likely to be under normal operation?

- a) Zone 21
- b) Zone 22
- c) Zone 1
- d) Zone 2

Q59 You are part of a team development a design for a highly complex process to be added to an existing major petrochemical plant. You will be leading the risk assessment.

What level of risk assessment is appropriate for this project?

- a) A qualitative risk assessment
- b) A semi-quantitative risk assessment
- c) A dynamic risk assessment
- d) A fully quantitative risk assessment

Q60 A chemical reaction process to make 4-tonnes of products is carried out once per month by mixing equal quantities of two chemicals. All chemicals are stored in storage tanks, each with a capacity of 60-tonnes. On one occasion, process water was accidentally allowed to enter one of the storage tanks, leading to an explosion and large scale release of gas and vapour.

Which of the following measures would have been the most inherently safe methods of reducing the potential consequences?

- a) Proper labelling of each tank
- b) Detailed process operating procedures
- c) Minimizing inventories stored
- d) Stored in ventilated buildings

Q61 You are responsible for ensuring safe isolation of a section of a chemical plant ahead of intrusive inspection and maintenance.

The highest standard of isolation that you can achieve is

- a) insertion of a blank plate.
- b) double block and bleed.
- c) Single valve isolation.
- d) double valve isolation.

Q62 A vessel is being pressure tested. The vessel is sealed off and water is pumped in. The water pressure is slowly increased to 150% of designed operational pressure and vessel is monitored to see if any leaks are visible .

What type of pressure testing does this describe ?

- a) Hydrodynamic
- b) Hydraulic
- c) Hydrostatic
- d) Hydropneumatic

Q63 During a major incident the primary plant audible warning system was activated, but management subsequently issued an instruction to switch it off shortly after.

Switching the primary plant audible warning system off in these circumstances was primarily a

- a) necessity to prevent noise pollution.
- b) decision to reduce nuisance alarms.
- c) chance to test back-up warning systems.
- d) deviation from a standard operating procedure.

Q64 several workers were killed by fire and explosion after a worker cut into the wrong pipe. The pipe contained flammable hydrocarbons and had not been isolated. The release was ignited by the sparks generated during the cutting operation.

You are the process safety representative investigating the accident. What is the most significant cause of this incident?

- a) Failure to control ignition sources
- b) Lack of deluge system
- c) Failure to control work
- d) Plant and process design

Q65 Which of the following is an example of a process safety risk reduction measure ?

- a) Wearing protective gloves when handling acidic waste.
- b) Safety signs that help ensure handrails are used on stairways.
- c) Building a brick bund wall around an acid storage tank.
- d) Safety rules that help ensure high-visibility vests are worn.

Q66 Competence at work is often demonstrated through the application of skilled behaviour based on sound knowledge gained through

- a) work experience**
- b) e-learning assessment**
- c) site audits**
- d) staff surveys**

Q67 Process safety performance indicators should contains a mix of both leading and lagging indicators.

An example of a leading indicator is the number of

- a) equipment inspections.**
- b) overpressure incidents.**
- c) Lost time injuries.**
- d) Loss of containments.**

Q68 The key elements of a process safety management system follow the Plan Do Check Act model.

'policy' aligns with which part of the model?

- a) Plan**
- b) Do**
- c) Check**
- d) Act**

Q69 which of the following is a lagging indicator ?

- a) Percentage of safety critical equipment successful tests**
- b) Number of safety critical equipment tests**
- c) Percentage of safety critical equipment failures**
- d) Number of safety critical equipment inspections**

Q70 shift handover between an incoming and outgoing operator should be conducted safely with

- a) at least one participant taking sole responsibility.**
- b) none of the participants taking joints responsibility.**
- c) both participants taking joint responsibility**
- d) an operations supervisor taking sole responsibility.**

Q71 Inspecting a workplace where contract workers are carrying out work is

- a) actively monitoring.
- b) reactively monitoring.
- c) condition monitoring.
- d) performance monitoring.

Q72 Inerting ensures that the concentration of oxygen is

- a) above the level that supports combustion in a flammable atmosphere.
- b) at the level that supports combustion in a flammable atmosphere.
- c) below the level that supports combustion in a flammable atmosphere.
- d) at or above the level that supports combustion in a flammable atmosphere.

Q73 Under What circumstances does brittle fracture occur?

- a) Form exposing materials to rapid and extreme temperature changes with uneven expansion and stress overcoming materials strength
- b) As a result of a repeated application of loads that, individually, do not create sufficient stress to cause failure
- c) With the gradual extension of materials under a steady tensile stress over a long period of high temperature conditions nearing the melting point of the material
- d) Suddenly when a material with no elasticity is put under excessive stress, or one with limited elasticity has no time to be elastic under excessive stress

Q74 What is the most likely fire risk presented by external floating roof tanks?

- a) Ignition of flammable vapour within the tank
- b) Impact with tankers igniting the contents due to friction
- c) Ignition of flammable vapour leaking past a defective rim seal
- d) Ignition of flammable vapour passing out of pressure vacuum relief valves

Q75 To manage an organisation's reputation during emergencies, there should be a nominated trained person to specifically liaise with

- a) Plant workers.
- b) Media reporters.
- c) Site visitors.
- d) Corporate managers.

Q76 The most effective way to familiarise people with their roles and responsibilities when faced with an emergency is a

- a) presentation**
- b) workshop**
- c) role play exercise**
- d) practical drill**

Q77 A large process vessel requires cleaning. This usually involves the workers entering the vessel following effective isolation. The visibility within the vessel is limited even with the provision of temporary lighting. Internal scaffolding is in place and supports the worker when jetting the internal surfaces. The jetting equipment operates at extremely high pressure.

Which of the following options should be considered FIRST to make the job safer ?

- a) Substitute the jetting machine for a lower pressure machine still capable of removing solid material from the contaminated surface.**
- b) Use a permit-to-work system for this high risk task to formally control workers carrying out the jetting activity.**
- c) Suitable guarding on the jetting equipment to prevent injuries from high pressure splashing and dislodged residues.**
- d) Provision of an automatic mechanical jetting machine instead of the requirement for vessel entry by a worker.**

Q78 You are trying to convince your Board to adopt a safety management system. You are using the argument that a safety management system can help to avoid needless and preventable deaths and injuries.

What type of argument is this usually known as?

- a) Morale**
- b) Medical**
- c) Municipal**
- d) Moral**

Q79 A chemical plant, located four kilometers from a city center, accidentally released a very large quantity of a toxic gas. The release of toxic gas occurred when a process operator mistakenly added water to a storage tank containing a toxic liquid. The two reacted violently causing the contents to boil.

The reaction that occurred within the storage tank was one that

- a) absorbed energy with a corresponding temperature drop.
- b) released energy with a corresponding temperature increase.
- c) released energy with a corresponding temperature decrease.
- d) absorbed energy with a corresponding temperature increase.

Q80 A significant plant change should be approved by

- a) expert accountants.
- b) senior management.
- c) practical consultants.
- d) department supervisor.

Q81 Topics of discussion between a contractor and permit issuer within a cold work permit would include

- a) supervision of the task and fire watchers required.
- b) duration of the task and gas testing requirements.
- c) location of the task and cancellation requirements.
- d) suspension of the task and fire blankets required.

Q82 Hot work permits are typically used where work could include

- a) generation of arcs, sparks or flames.
- b) high voltage (HV) electricity cables.
- c) emergency and rescue procedures.
- d) isolation of hot rotating machinery.

Q83 Which of the following is a significant process safety risk from the increased use of contractors?

- a) Detailed knowledge of equipment and processes may be lost when contractors change.
- b) Contractors are unfamiliar with the site and are more likely to have incidents.
- c) Selection of contractors uses resources that could be used improving process safety.
- d) Contractors are not always available when needed and can be unreliable.

Q84 It's very common practice to segregate an accommodation block from process plant.

Which type of risk control is this an example of?

- a) Elimination
- b) Engineering control
- c) Administrative control
- d) Substitution

Q85 Liquid is being transferred from a road tanker into a storage vessel.

The failure to open a vent valve on the tanker during the transfer could result in a tanker

- a) explosion.
- b) overflow.
- c) over-pressure.
- d) implosion.

Q86 Alerting and informing when problems occur during process operation is the main function of

- a) an indicator.
- b) a dial.
- c) an alarm.
- d) A gauge.

Q87 A thorough visual inspection of the plant to determine that all work has been completed is most relevant within the

- a) Post-start-up safety review.
- b) Pre-shutdown safety review.
- c) Post-shutdown safety review.
- d) Pre-start-up safety review.

Q88 In order for a process plant to start-up safely PRVs must be in working order.

PVRs are

- a) Process resource values.
- b) Pressure relief valves.
- c) Pressure resistance valves.
- d) Process running valves.

Q89 Overfilling of tanks can be prevented most effectively through a combination of

- a) Calculation of mass flow rates in, knowledge of tank capacity and low level alarm.
- b) Calculation of mass flow rates in and mass flow rates out, knowledge of tank capacity and low level alarms.
- c) Calculation of mass flow rates in and mass flow rates out, knowledge of tank capacity and high level alarms.
- d) Calculation of mass flow rates out, knowledge of tank capacity and high level alarm.

Q90 Which of the following properties of dust are required for dust explosion to occur?

- a) High humidity and moisture content.
- b) Dispersion and confinement of dust.
- c) Relatively large dust particle size.
- d) Dust cloud in a relatively open area.

Q91 At storage depots, flammable liquids may be pumped between storage vessels, pipelines and tankers.

Which of the following measures is most effective in preventing flammable vapour releases?

- a) Maintaining plant integrity.
- b) Vapour detection systems.
- c) Emergency pump shutdown.
- d) Storage tank bunds.

Q92 One of the main requirements for a primary dust explosion is occur is a combustible dust that is

- a) airborne and within explosive limits.

- b) Airborne and outside explosive limits.
- c) on a hot flat surface outside explosive limits.
- d) on a hot flat surface within explosive limits.

Q93 The transfer of heat by the movement of hot fluids is known as

- a) conduction.
- b) convection.
- c) radiation.
- d) diffusion.

Q94 How does an intumescent material operate in a fire?

- a) Contracts when heated to prevent the spread of fire and smoke
- b) Expands when heated to prevent the spread of fire and smoke
- c) Illuminates when heated to help evacuation in the event of a fire
- d) Radiates when heated to increase the spread of fire and smoke

Q95 A drilling rig has been on site for a number of weeks drilling offshore. Contractors have now completed the drilling of the well and the well appears to have been made safe. A group of senior managers, from the large corporation that own the well, fly in by helicopter to visit at a safety critical stage.

What should the visitors be taking notice of at this crucial stage, to prevent a major incident?

- a) How much money the drilling activity is costing.
- b) The condition of the platform given the harsh environment.
- c) Can savings be made by more effective working practices.
- d) Process operations going on at the time of the visit.

Q96 A facility stores petroleum (gasoline) and liquefied petroleum gas (LPG) in bulk storage vessels. The storage vessels are being filled from a pipeline when the connecting pipe between two of the storage vessels ruptures.

A violent explosion results after the vapour cloud finds an ignition source in a remote part of the site. For the following 90 minutes a series of further explosions occur with associated missiles as each of the storage vessels explodes in turn.

- a) Effective plant layout with respect to positioning of bulk storage vessels.

- b) Suitable site emergency planning with good emergency vehicle access.
- c) Suitable active and passive fire protection to prevent fires from spreading.
- d) Effective inspection and maintenance programme for plant pipework.

Q97 You are the process safety lead during the design phase of a new continuous chemical reactor. You are interested in identifying both hazards from the process potential improvements to operation of the reactor to increase efficiency and reduce downtime.

The technique best designed to help meet these aims is

- 1. Failure mode effects analysis.
- 2. Hazard and operability study.
- 3. Event tree analysis.
- 4. What-if study.

Q98 In accident investigation, an unsafe act or unsafe condition that directly leads to damage is known as an

- a) immediate cause
- b) underling cause
- c) unlikely cause
- d) incidental cause

Q99 Health and safety committee meetings mainly promote

- a) compliance
- b) co-ordination
- c) consultation
- d) compartmentation

Q100 For which of the following fire precautions are inert gases most suitable?

- a) Automatic fire detection
- b) Fire prevention
- c) Fire suppression
- d) Passive fire protection

Q101 A large proportion of hydrocarbon storage tank fires are due to lightning strikes.

Which of the following types of storage tanks is most vulnerable to fires from this cause?

- a) Fixed roof
- b) Floating roof
- c) Cone roof
- d) Umbrella roof

Q102 Maintenance work is planned on Plant A. The work is likely to have a significant impact on the operation of a second plant, plant B.

What is the most effective way of controlling risk between the two plants in this situation?

- a) Display a warning sign between plants A and B
- b) Display a single permit-to-work in plant A only
- c) Display a single permit-to-work in plant B only
- d) Display a duplicate permit-to-work in plants A and B

Q103 A process line can be isolated by closing two-in-line valves and opening a drain between them.

which type of isolation does this describe?

- a) Single valves isolation
- b) Double block and bleed
- c) Control valve isolation
- d) Single block and bleed

Q104 Sometimes a 'table-top-exercise' is used to simulate an emergency.

What are the main benefits of such exercises?

- a) Validates plans, develops staff competency by practising roles and tests procedures
- b) Identifies hazards, considers those affected, evaluates risk and implements suitable controls
- c) Analyses unsafe working conditions and behaviours and implements corrective actions
- d) Examines a process safety management system, identifies problems and provides solutions

Q105 One of the main typical features of a jet fire is

- a) a highly directional flame.
- b) a rapid spread in every direction.
- c) a flame with an enormous surface area.
- d) A diffuse flame with low momentum.

Q106 Which type of fire extinguishing medium works by the mechanisms of suffocation and vapour suppression?

- a) Powder
- b) Foam
- c) Inert
- d) Water

Q107 Which of the following is the major hazard from layers of dust accumulating in the plant and on machinery?

- a) Plant corrosion leading to a major leak.
- b) Secondary dust explosions.
- c) Primary dust explosions.
- d) Creation of static electrical ignition.

Q108 When filling a vessel with a flammable liquid, one of the main factors that increases the generation of an electrostatic charge is

- a) overfilling.
- b) splash filling.
- c) bottom filling.
- d) continuous filling.

Q109 An accident report detailed that similar plants owned by the same organisation in other international locations were built downwind of cities.

This was as a result of previous incidents where smaller quantities of toxic gas and had been released into the atmosphere. This chemical plant was built upwind of a city.

This is primarily an example of a failure in

- a) operational control.
- b) asset management.
- c) management of change.
- d) organizational learning.

Q110 There has been a near miss incident where an experienced operator failed to respond to a high level alarm on a fuel tank, resulting in a loss of containment from the tank. The incident could have had major catastrophic consequences had the fuel found an ignition source. It is important that lessons are learnt from all incidents, including near misses.

Which of the following is the best way to learn lessons from this incident and prevent such an incident happening again?

- a) Discuss the incident with the operator as a personal matter. Ensure that they understand the significance of the incident and that they must respond appropriately to alarms in the future

- b) Investigate the incident thoroughly ,identify the root causes and ensure this is communicated to as wide an audience as possible
- c) Accept that on this occasion a mistake was made. Offer the operator refresher training that reminds them of the need to manage alarms effectively
- d) Suggest the organisation re-train the operator for a new role where they will not be put in charge of alarm management

Q111 A large storage tank had regularly been filled with liquid A. During refilling , a mistake occurred and a higher density liquid B was pumped into the empty tank instead. When the tank was only 75% full, it began to deform at the base. The tank had been well-maintained and no previous incidents of this type had been recorded with liquid A.

What is likely to be the most significant cause of the tank deformation?

- a) The weight of the liquid exceeds the design load.
- b) Liquid was added to the tank too quickly
- c) The tank was filled beyond what it was designed to hold
- d) The tank base was not properly anchored

Q112 A large storage tank, located above ground, is being filled with petroleum (gasoline) . The tank eventually overflows and petroleum cascades to the ground.

The subsequent vapour could spread and contact an ignition source located a considerable distance away. This initially results in a flash fire but is quickly followed by an explosion and series of fires engulfing the whole site.

What is most likely to have prevented this incident?

- a) Hazardous area classification and selection of suitable ignition-protected electrical and mechanical equipment.
- b) A high level alarm with an independent high level trip system and an automatic over protection system.
- c) Constant flow filling rate combined with accurate calculations of tank filling based manual level readings.
- d) Local fire departments having sufficient training and resources to respond to induction fires and explosions.

